The Impact of Noise on Pan-Tompkins QRS Detection Performance

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***Abstract*—This paper analyses the impact of baseline wander noise on the detection of the QRS Complex in ECG signals using the Pan-Tompkins algorithm under varying amounts of noise. Also investigated is the performance of an IIR Notch Filter at removing baseline wander, assessed as the QRS detection performance at different cut-off frequencies.**

***Index Terms*—QRS Detection, Pan-Tompkins Algorithm, ECG Signal Processing, IIR Filters, Baseline Wander**

I. Introduction

An ECG is an important tool for monitoring cardiac activity and diagnosing cardiac events. Detection of the QRS Complex is a key step in pipelines for automated ECG signal analysis. The Pan-Tompkins algorithm is a well-established method for detecting the QRS complex in real-time. In real-world ECGs, the signals are rarely clean, with respiration, electrical signals from muscles, and electrical mains hum introducing noise into the signal. Baseline Wander is the most common source of noise in an ECG signal, arising from the patient breathing or moving slightly.

This paper investigates how baseline wander affects the performance of the Pan-Tompkins algorithm across a range of signal-to-noise ratios, and examines whether a simple IIR notch filter can mitigate the effects of baseline wander noise to an acceptable level for QRS Detection. Three filter configurations were tested by varying the pole radius to control the cut-off frequency.

II. Test Conditions

Baseline Wander (BW) noise was loaded from the PhysioNet dataset which contains a clean ECG signal alongside extracted baseline wander noise. The BW noise was then applied to the ECG signal at varying Signal-to-Noise ratios, starting at 40dB, then continuing from 0dB to -36dB in steps of -3 dB. 50% of the dataset was used for brevity (c. 325,000 samples), and a tolerance of 50ms was allowed when running QRS detection to ensure detections could be properly categorised. The BW noise signal was scaled to the required power as follows:

$$P_{scaled} = \frac{P_n}{10^{\frac{SNR}{10}}} \tag{1}$$

The scaling factor K can then be calculated as:

$$K = \sqrt{\frac{P_{scaled}}{P_n}} \tag{2}$$

The Pan-Tompkins QRS Detection Algorithm was then applied to the noisy signal, initially with no filtering applied, followed by IIR notch (band-stop) filters designed with values of $r = 0.99$, $r = 0.96$, and $r = 0.93$ as follows:

$$H(z) = \frac{1 - z^{-1}1}{1 - rz^{-1}1} \tag{3}$$

For each run of the Pan-Tompkins algorithm, the True Positive, False Positive, and False Negative counts were recorded, along with Sensitivity, Precision, and Data Error Rate calculations. The Data Error Rate (DER) was calculated as follows:

$$DER = \frac{FP + FN}{TP + FN} * 100 \tag{4}$$

This allowed for the assessment of the impact of baseline wander noise on the algorithm, and additionally, a study of the effectiveness of IIR notch filters for removing this type of noise.

III. Results

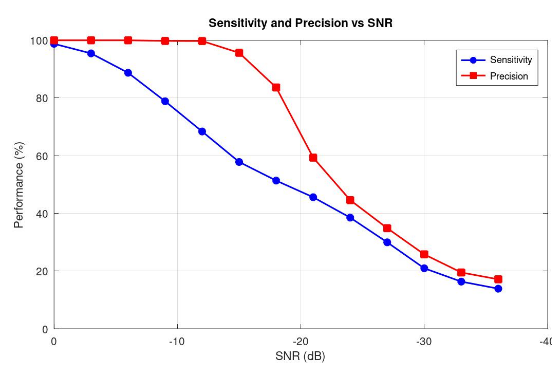

Fig. 1: QRS Detection Performance at a range of SNRs (No filtering)

The graph in Figure 1 shows how any amount of noise in the ECG signal can have a performance impact. The effects first begin to show in Sensitivity performance, which is a measure of true positives against all known positives in the dataset - in a good system, it effectively reflects the false negative rate, which is the behaviour seen here. Table I shows that at

0 dB, False Negatives are already beginning to creep in at a rate of 1.22%. This becomes noticeably worse as the SNR decreases with nearly half of the beats being missed altogether by -15dB. The precision performance, which reflects the ability to minimize false positives without misclassifying true positives, remains noticeably steady compared to Sensitivity, until around -15dB where there is a clear ‘cliff-edge’ after which performance takes a nose-dive. After -15dB, Table I shows how the false positive rate skyrockets due to the amount of noise in the signal. With low amounts of noise present in the signal, the main issue is beats not being strong enough to be detected, rather than noise being detected as real beats. By the time the noise is strong enough to introduce false beats, the signal is already gone far beyond useful.

TABLE I: QRS Detection Performance (No Filter)

SNR (dB)	TP	FP	FN	Sens (%)	Prec (%)	DER (%)
40	1145	0	0	100	100	0
0	1131	0	14	98.78	100	1.22
-3	1093	0	52	95.46	100	4.54
-6	1016	0	129	88.73	100	11.27
-9	903	2	242	78.86	99.78	21.31
-12	783	2	362	68.38	99.75	31.79
-15	662	30	483	57.82	95.66	44.8
-18	588	115	557	51.35	83.64	58.69
-21	522	358	623	45.59	59.32	85.68
-24	441	548	704	38.52	44.59	109.34
-27	343	641	802	29.96	34.86	126.03
-30	240	691	905	20.96	25.78	139.39
-33	187	772	958	16.33	19.5	151.09
-36	159	769	986	13.89	17.13	153.28

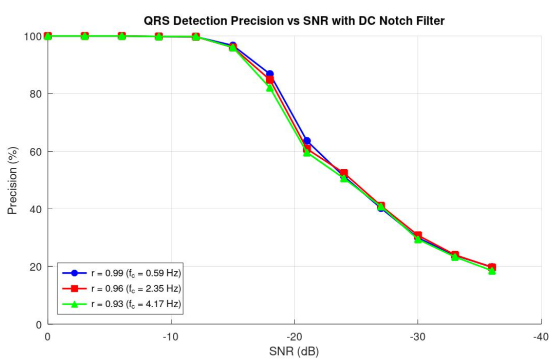

Fig. 2: QRS Detection Precision at a range of SNR (Notch filter at varying cut-off frequencies)

Introducing an IIR notch filter to remove the baseline wander noise, the precision performance continues to behave almost identically - to the point where there is still a noticeable cliff-edge at -15dB. This doesn’t change even as the cut-off frequency becomes 8x as larger than the initial filter parameters. This largely makes sense, as precision is effected mainly when the noise becomes so strong as to start introducing false beat detections. The filter is designed to prevent false negatives at reasonable SNRs rather than suppress false positives at arbitrarily low SNRs.

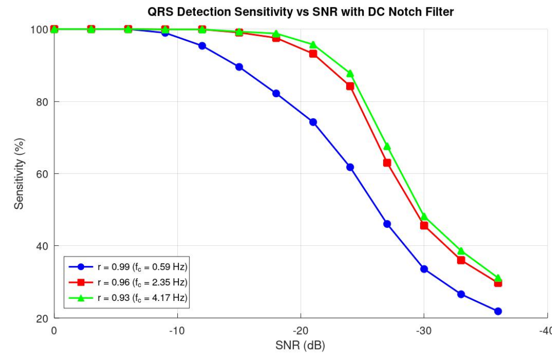

Fig. 3: QRS Detection Sensitivity at a range of SNRs (Notch filter at varying cut-off frequencies)

Returning to sensitivity further backs this theory, clearly showing that Sensitivity performance (i.e. FN rate) is significantly improved with proper filtering in place, with Figure 3 demonstrating that a cut-off frequency between 2 and 4 Hz is most effective at removing baseline wander. Detection performance remains highly accurate at as low as -21dB before facing the performance cliff with values of $r = 0.96$ and $r = 0.93$. While a value of $r = 0.99$ does show much improvement over no filter at all, it leaves significant performance gains on the table compared to higher values.

TABLE II: QRS Detection Performance (Notch Filter, $r = 0.99$)

SNR (dB)	TP	FP	FN	Sens (%)	Prec (%)	DER (%)
40	1145	0	0	100	100	0
0	1145	0	0	100	100	0
-3	1145	0	0	100	100	0
-6	1145	0	0	100	100	0
-9	1133	2	12	98.95	99.82	1.22
-12	1092	3	53	95.37	99.73	4.89
-15	1025	35	120	89.52	96.7	13.54
-18	941	143	204	82.18	86.81	30.31
-21	850	487	295	74.24	63.58	68.3
-24	707	673	438	61.75	51.23	97.03
-27	527	783	618	46.03	40.23	122.36
-30	384	900	761	33.54	29.91	145.07
-33	304	971	841	26.55	23.84	158.25
-36	250	1014	895	21.83	19.78	166.72

IV. Discussion & Conclusion

Removing the 40dB SNR results from the graph allows a much closer look at the effects each -3dB jump has on the precision and sensitivity characteristics of the algorithm. Specifically, this makes the gap between $r = 0.96$ and $r = 0.93$ a bit clearer, highlighting how little difference there is between the two. Looking into the numbers for both, there is evidence to suggest that $r = 0.96$ is actually worse than $r = 0.93$. Beyond -15dB, the $r = 0.93$ filter actually has a higher False Positive Rate across the board, despite fewer False Negatives, even though the number of True Positive detections actually increases - this suggests that ‘r=0.93‘ may actually be

TABLE III: QRS Detection Performance
(Notch Filter, $r = 0.96$)

SNR (dB)	TP	FP	FN	Sens (%)	Prec (%)	DER (%)
40	1145	0	0	100	100	0
0	1145	0	0	100	100	0
-3	1145	0	0	100	100	0
-6	1145	0	0	100	100	0
-9	1144	2	1	99.91	99.83	0.26
-12	1144	3	1	99.91	99.74	0.35
-15	1134	47	11	99.04	96.02	5.07
-18	1117	200	28	97.55	84.81	19.91
-21	1067	688	78	93.19	60.8	66.9
-24	964	875	181	84.19	52.42	92.23
-27	721	1035	424	62.97	41.06	127.42
-30	522	1174	623	45.59	30.78	156.94
-33	412	1305	733	35.98	24	177.99
-36	340	1388	805	29.69	19.68	191.53

TABLE IV: QRS Detection Performance
(Notch Filter, $r = 0.93$)

SNR (dB)	TP	FP	FN	Sens (%)	Prec (%)	DER (%)
40	1145	0	0	100	100	0
0	1145	0	0	100	100	0
-3	1145	0	0	100	100	0
-6	1145	0	0	100	100	0
-9	1144	2	1	99.91	99.83	0.26
-12	1144	2	1	99.91	99.83	0.26
-15	1137	47	8	99.3	96.03	4.8
-18	1131	248	14	98.78	82.02	22.88
-21	1096	745	49	95.72	59.53	69.34
-24	1005	982	140	87.77	50.58	97.99
-27	774	1122	371	67.6	40.82	130.39
-30	551	1322	594	48.12	29.42	167.34
-33	442	1455	703	38.6	23.3	188.47
-36	356	1567	789	31.09	18.51	205.76

too low a radius and that the true ‘best filter’ could potentially exist at values even higher than $r = 0.96$, but certainly above $r = 0.93$. The data error rate adds further support to this theory, as the DER actually increases significantly over $r = 0.96$ at SNRs below -18dB.

Overall, it is clear that noise can have significant impact on QRS Detection performance at low SNRs, but also that filtering can clear up the majority of this noise to an acceptable level for heartbeat detection. With precision performance, regardless of the presence of a filter, there is a clear falling off point in the graph when the amount of noise gets high enough to cause false beats to be detected. Sensitivity doesn’t show quite the same level of harsh falloff, acting closer to linear at lower SNRs, but with a significant falloff after -15dB.